

LUKE AFB, AZ, USA

WMO: 722785

Lat: 33.533N

Lon: 112.383W

Elev: 1085

StdP: 14.13

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23111

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	35.0	37.6	5.1	7.4	64.0	9.4	9.3	65.3	23.0	56.5	18.7	54.1	5.4	340	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	111.0	69.7	108.7	69.6	106.6	69.4	76.1	92.3	75.2	93.1	74.4	93.6	11.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.7	126.4	81.0	70.7	117.6	82.8	68.6	109.4	85.0	40.2	92.0	39.2	92.1	38.5	93.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.7	18.1	16.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	30.4	115.8	3.2	2.7	28.1	117.7	26.3	119.3	24.5	120.8	22.2	122.7	
			WB	26.3	78.4	3.5	1.6	23.8	79.5	21.7	80.5	19.7	81.4	17.2	82.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	54.0	57.3	64.3	70.3	78.3	88.5	94.0	93.1	86.6	73.8	62.6	53.3
	DBStd	15.44	5.14	6.10	5.71	6.46	6.24	5.56	4.19	4.16	4.99	6.26	5.99	5.31
	HDD50	62	22	10	0	0	0	0	0	0	0	0	2	28
	HDD65	1157	340	220	84	24	1	0	0	0	0	11	113	364
	CDD50	8488	147	215	442	609	877	1156	1364	1336	1097	738	378	129
	CDD65	4108	0	5	61	183	413	706	899	871	647	283	40	0
	CDH74	64872	25	175	1056	2569	5947	11863	15286	14518	9469	3368	581	15
	CDH80	41835	0	18	334	1154	3383	8139	10928	10180	5910	1645	144	0
Wind	WSAvg	6.8	5.6	6.0	6.5	7.9	7.9	7.7	8.3	7.5	6.7	6.1	5.6	5.2
Precipitation	PrecAvg	8.3	1.0	1.0	0.9	0.3	0.1	0.0	0.8	1.3	0.7	0.6	0.6	1.1
	PrecMax	17.8	6.6	4.6	3.1	1.6	1.3	0.4	2.5	2.9	2.9	3.7	2.5	4.0
	PrecMin	4.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.3	1.5	1.3	0.7	0.4	0.3	0.1	0.7	0.7	0.7	0.7	0.6	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	82.7	91.4	98.0	105.2	115.0	114.5	112.9	107.6	99.7	88.5	76.0
		MCWB	52.6	55.5	58.5	60.8	63.6	69.3	70.1	70.5	70.1	65.8	58.7	54.2
	2%	DB	73.2	79.1	86.8	94.2	101.2	109.8	111.1	109.7	104.8	95.4	84.1	72.4
		MCWB	51.5	54.8	56.9	59.5	62.1	66.8	70.1	71.3	68.9	63.2	57.3	52.7
	5%	DB	70.2	75.6	83.8	90.7	98.3	107.3	108.8	107.8	102.4	92.6	80.7	69.0
		MCWB	50.7	53.9	55.8	58.2	61.5	66.1	70.3	71.3	68.3	61.8	56.5	51.5
	10%	DB	66.8	71.9	80.4	87.1	95.2	104.7	106.5	105.4	100.0	89.5	77.1	65.5
		MCWB	49.7	52.6	54.8	56.8	60.6	65.1	70.4	71.4	68.7	61.2	55.5	50.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	60.0	60.8	63.2	66.3	73.2	77.3	77.6	76.2	71.8	62.0	59.4
		MCDB	64.5	72.7	82.0	91.6	89.8	104.1	88.7	93.7	87.0	82.6	75.6	65.7
	2%	WB	56.2	57.8	59.3	60.8	64.4	71.1	75.9	76.3	74.8	67.9	60.1	57.1
		MCDB	62.7	70.9	78.8	89.2	93.6	100.7	91.7	94.0	89.4	85.8	76.5	65.0
	5%	WB	53.7	56.1	57.6	59.3	63.1	69.1	75.1	75.3	73.5	65.7	58.4	55.0
		MCDB	63.3	69.5	76.4	86.2	93.4	99.6	92.9	94.5	90.7	84.1	75.7	63.2
	10%	WB	51.7	54.4	56.2	57.9	61.6	67.2	74.3	74.5	72.5	63.7	56.8	52.8
		MCDB	62.8	68.2	75.7	84.2	91.8	100.1	93.7	94.6	90.5	83.6	74.2	62.0
Mean Daily Temperature Range	5% DB	MDBR	24.1	24.8	27.7	28.0	27.7	29.4	23.7	23.0	24.5	27.3	26.2	23.0
		MCDBR	29.4	30.6	32.8	32.9	32.1	31.6	27.0	26.1	27.3	30.9	30.6	27.7
	5% WB	MCWBR	13.6	13.1	11.7	11.7	10.3	9.7	7.5	6.9	8.0	11.4	13.0	13.2
		MCDBR	21.1	25.5	28.5	29.4	28.5	26.6	21.1	21.4	20.2	23.9	25.9	20.8
Clear-Sky Solar Irradiance	taub	0.281	0.286	0.311	0.325	0.335	0.344	0.403	0.392	0.348	0.319	0.289	0.280	
	taud	2.567	2.561	2.468	2.412	2.413	2.389	2.351	2.375	2.468	2.505	2.547	2.571	
	Ebn at Noon	301	310	307	305	301	297	279	281	290	292	294	294	
	Edh at Noon	25	28	33	37	37	38	39	38	33	29	25	24	
All-Sky Solar Radiation	RadAvg	1075	1381	1852	2305	2582	2677	2356	2164	1944	1559	1189	973	
	RadStd	77	102	84	62	80	95	104	69	68	85	47	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (13 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	-149	+193	+68

Nomenclature: See separate page